

Order-Service <-> Inventory-Service gRPC Guide

Purpose: quick refresher on how create-order checks stock via gRPC, with file map, call sequence, and commands.

1) Big Picture (Simple)

- 1 Client calls **POST /api/orders** on **order-service**.
- 2 **OrderService** calls gRPC client before saving.
- 3 gRPC request goes to **inventory-service** method **CheckStock**.
- 4 Inventory gRPC handler calls **InventoryService.isInStock(productId, quantity)**.
- 5 gRPC response returns only **in_stock = true/false**.
- 6 Order is saved only when **in_stock** is true; otherwise order-service returns conflict.

2) Files You Need

inventory-service (gRPC server)

- inventory-service/src/main/proto/inventory.proto - gRPC contract (service/method/messages).
- inventory-service/pom.xml - gRPC/protobuf dependencies and protobuf-maven-plugin.
- inventory-service/src/main/java/.../grpc/InventoryStockGrpcService.java - implements checkStock(...).
- inventory-service/src/main/java/.../config/GrpcServerConfiguration.java - starts gRPC server and binds service.
- inventory-service/src/main/resources/application.properties - grpc.server.port=9092.

order-service (gRPC client)

- order-service/src/main/proto/inventory.proto - same contract (or shared later).
- order-service/pom.xml - gRPC/protobuf dependencies and protobuf plugin.
- order-service/src/main/java/.../client/InventoryGrpcClient.java - creates channel + blocking stub + isInStock call.
- order-service/src/main/java/.../service/OrderService.java - calls gRPC before save.
- order-service/src/main/resources/application.properties - inventory.grpc.host and inventory.grpc.port.

3) How Routing Works

- 1 The route is defined in **inventory.proto**: service InventoryStockService and rpc CheckStock(...).
- 2 Maven compile generates Java classes into target/generated-sources/protobuf/...
- 3 Server class extends generated base: InventoryStockServiceGrpc.InventoryStockServiceImplBase.
- 4 When client calls CheckStock, gRPC runtime dispatches to overridden checkStock(...) method.

4) Build/Generate Sequence

Run in this order:

```
cd /home/hassan/D/work/springboot/MicroServices/inventory-service ./mvnw clean compile cd /home/hassan/D/work/springboot/MicroServices/order-service ./mvnw clean compile
```

5) Runtime Sequence (Who Calls Whom)

- 1 Start DB containers for both services.
- 2 Start inventory-service first (gRPC port 9092 available).

- 3 Start order-service.
- 4 Send POST /api/orders to order-service.
- 5 order-service -> InventoryGrpcClient.isInStock -> gRPC CheckStock.
- 6 inventory-service evaluates stock and returns in_stock.
- 7 order-service saves or rejects order based on boolean result.

6) Quick Test Commands

Start inventory-service:

```
cd /home/hassan/D/work/springboot/MicroServices/inventory-service docker compose up -d  
./mvnw spring-boot:run
```

Start order-service:

```
cd /home/hassan/D/work/springboot/MicroServices/order-service docker compose up -d ./mvnw  
spring-boot:run
```

Seed inventory and test create-order:

```
curl -X PUT http://localhost:8082/api/inventory -H "Content-Type: application/json" -d  
{"productId":"p1","quantity":3} curl -i -X POST http://localhost:8081/api/orders -H  
"Content-Type: application/json" -d  
{"productId":"p1","price":100.00,"sku":"sku1","quantity":2} curl -i -X POST  
http://localhost:8081/api/orders -H "Content-Type: application/json" -d  
{"productId":"p1","price":100.00,"sku":"sku1","quantity":5}
```

Cheat Sheet (1 Page)

- Contract file: src/main/proto/inventory.proto
- Generate classes: ./mvnw clean compile
- Generated paths: target/generated-sources/protobuf/java and grpc-java
- Do not edit generated files in target/
- Server impl: InventoryStockGrpcService.checkStock(...)
- Server bootstrap: GrpcServerConfiguration (port 9092)
- Client call: InventoryGrpcClient.isInStock(productId, quantity)
- Order decision: OrderService.placeOrder(...) -> save only when true
- Common issue: red imports -> reload Maven project + recompile
- Manual sanity check: seed inventory then POST /api/orders

Call chain memory trick: **REST -> OrderService -> gRPC Client -> gRPC Server -> InventoryService -> boolean -> OrderService decision**